A Short-Range Infrared Communication for Swarm Mobile Robots

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***Abstract*— This paper presents another short-range communication technique suitable for swarm mobile robots application. Infrared is used for transmitting and receiving data packets and obstacle detection. The infrared communication system is used for an autonomous mobile robot (UPM-AMR) that will be used as a low-cost platform for robotics research. A pulse-code modulation (PCM) digital scheme is used for transmitting data. The reflected infrared signal is also used for distance estimation for obstacle avoidance. Analysis of robot's behaviors shows the feasibility of using infrared signals to obtain a reliable local communication between swarm mobile robots.**

Keywords- ***Infrared; Communication; PCM; Swarm; Mobile robots;***

I. INTRODUCTION

Robots are being utilized increasingly in working tasks around humans. Robots are used in many applications such as office, military tasks, hospital operations, industrial automation, security systems, dangerous environment, and agriculture. Swarm mobile robot is a new coordination approach for multiple robots to cooperatively achieve a single task. Each of the mobile robots in the swarm should have autonomous behavior with out any central controller. Thus, communication between mobile robots is a significant task which allows multiple robots to accomplish complex behaviors in swarm robots' scenarios.

The autonomous mobile robots (AMRs) can not execute their programmed tasks without reliable data transmission techniques [1-4]. Mobile robots use various communication methods such as wireless network [5], Bluetooth [6], Ultrasonic and Infra-Red [7-8]. Each method has its own advantages and drawback for different AMR application scenarios. Wireless network and Bluetooth are suitable communication approach for outdoor applications that does not requires the information of distance and location of surrounding robots. Providing distance and location information with wireless communication is complicated and requires additional hardware and software to solve position estimation problem [9].

Infrared (IR) light is an electromagnetic radiation with longer wavelength than visible light. The IR wavelength is between 750nm and 1mm. It is used in many applications such as military, thermal efficiency, remote temperature sensing, and short-range wireless communication. The IR filters are used to block visible light spectrums which are employed in receiver's pack. The IR is divided into three bands [10]:

- IR-A: 700 nm-1400 nm
- IR-B: 1400 nm-3000 nm
- IR-C: 3000 nm-1 mm

The proposed hardware uses IR components with 950nm wavelength that are in IR-A band. This wavelength is the popular wavelength in many short-range remote controlling systems that employ IR wireless data communication. Several developed standards are used in IR data transmission [11] and are common approach in multiple-robots communication environment.

In this paper, we introduce a short-range communication technique based on PCM for autonomous mobile robots. All messages are modulated without complex algorithms. Some standards such as IrDA and AIr are defined for multi-channel users and they use reliable IR transmission algorithms but they are not suitable for our application that uses small size packets in short distance. Robot's messages are modulated with simple unique packet format and they are broadcasted in robotic environment by IR radiation. In addition, the IR signal reflects are used to detect surrounding obstacles and used to estimate the obstacle distance. The deployed robot is designed for indoor application and low-cost components are a requirement.

Multiple robotic environments are described in section II. Section III describes the requirements for implementing the proposed communication. The designed architecture is described in section IV. The experimental results are presented in section V.

II. MULTIPLE ROBOTIC ENVIRONMENT

In swarm robotic scenarios, all AMRs must be able to detect other surrounding robots in the environment. Each robot should be able to calculate relative position, velocity, orientation, and distance of other AMRs. The actual dynamic recognition of AMRs depends on the types and number of sensors that are employed. These AMRs execute global behavior based on the communication between each AMR.

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These behaviors require constant updating of sensor-based information between each AMR. The capability of the sensors and communication system is significant parameter for AMR's cooperation.

Various communication techniques have been evaluated in multiple robot environments. Several researchers use image processing technique for multiple AMRs environment recognition [12, 13]. Implementation of vision-based technique is complicated and requires a lot of computing resources. Radio communication has also been used for multiple AMRs environment [5, 6]. Although radio communication allow long distance communication, several other problem exist [14]. One of these problems is the limited communication channel especially when a large number of AMRs is deployed in the environment. Radio-based communication also faced with distance estimation and location approximation problems.

The local communication technique is the most appropriate method for distributed robotic systems [15]. In this work, infrared is used to implement a reliable local communication as well as sensor system. In the following section, the detail requirement of the UPM-AMR communication requirements will be presented.

III. AMR Communication Requirements

In this section the requirement of the IR component, robot controller and the communication will be elaborated. Several tests preformed shows that, the selected IR components as described in this section provide better results than others for this application. The controller module that is responsible transmission and receiving IR packets to between robots will be presented. The communication system responsible to format IR packets has been implemented with an optimal algorithm, necessary for real-time autonomous system.

A. IR components

The proposed technique uses two IR components; IR emitter and IR phototransistor as shown in Fig.1. These components are provided in small plastic package. Fig.1 (a) is a phototransistor with visible light spectrum filter and Fig.1 (b) is the IR emitter diode with side-view packet.

TEFT-4300 is a sensitive and high speed phototransistor with wide viewing angle feature of approximately 60º. It is suitable for sensing nearby IR radiation with fast response time of micro-second range. The maximum working frequency of this receiver is 180 kHz, which provides enough baud-rate to implement proposed local data transmission technique.

TSKS-5400 is a standard IR emitting diode in a side-view plastic package. A small recessed spherical lens provides an improved radiant intensity in a low profile case with peak wavelength 950nm. The maximum radiant power of this emitter component is approximately 10mW, with switching time is in millisecond range.

B. Robot controller

An ATMEGA168 microcontroller is used in implementation of the robot controller. This chip is employed as the AMR's main processor which manages the IR communication, decides on the task, memory management, and controlling motors simultaneously.

The robot main board is equipped with six IR diodes and six phototransistors to communicate between AMRs as shown in Fig. 2. This arrangement allows each AMR to communicate in any direction with other AMRs in its local vicinity. Each IO works in both digital and analog modes simultaneously providing fast communication with neighboring AMRs and reliable sensing of reflected IR radiation from obstacles. In addition, the feedback serial data is sent to PC to read the memory blocks and saved the sensors' absorbed values when it is requested. The architecture of AMR sensor board is shown in Fig. 3.

C. Communication

The AMRs will rely on IR to provide reliable short-range communication. The IR sensors will also be used for obstacle detection and distance estimation. Two types of communication modes are available depending on the importance of the message. These modes are communication with and without confirmation. The confirm-based communication mode provides reliable data transmission between AMRs. However, this mode requires additional computational resources compared to the other mode. The communication mode without confirmation or broadcast mode is suitable for high speed communication (i.e. collision reaction).

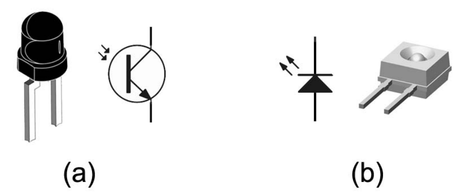

Figure 1. (a) IR phototransistor (TEFT4300), (b) IR emitter diode (TSKS5400)

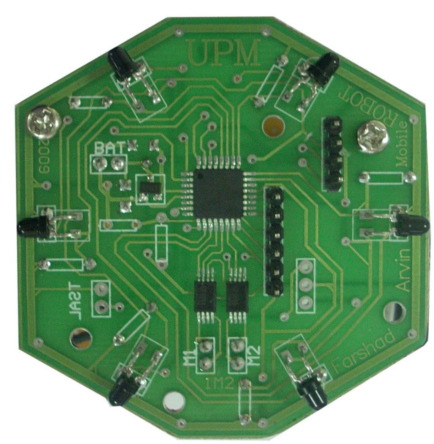

Figure 2. The main board of mobile robot (60º receivers' topology)

The size of data packet is an important parameter for reliable data communication. All communication data packet use similar packet format as shown in Fig. 4. Two types of instructions are available for use in the swarm scenarios, i) the basic control instructions such as trajectory control, obtain sensors reading and broadcast packets, ii) the high level instructions for implementing global behaviors which depend on swarm scenarios.

IV. Design of Reliable Infrared Transceivers

In implementation of communication systems for multiple AMR's cooperation and sensory system for ability to detect obstacles and other AMRs, some design decisions are discussed in this section. The AMR must be able to detect obstacles and other AMRs in dynamic robotic environment to prevent collision. The AMR should be able to detect the other AMR's trajectory information such as velocity, position, and moving direction. The AMRs are able to recognize surrounding mobile robots with their uniquely defined identification (ID) numbers. The communication and sensory systems should be small and low-power so that it can be operated by robot's battery.

A. *Sensory Module Design*

As the sensors are used for reactive behavior, the microcontroller will serve the input by its interrupt routine. The received analog signals are digitised by microcontroller's analog to digital converter (ADC) and used by the distance estimation function. The microcontroller would then call a function to decode the received IR packet. As the ID of the AMRs is included in the broadcast, the AMRs are able to detect other robots and obstacles as shown in Fig. 5. The AMR detects obstacles when it receives its reflected ID. Further IR packet digestion for communication purposes will be handled by the communication module.

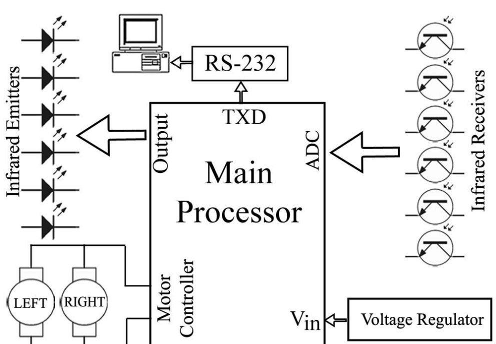

Figure 3. Architecture of proposed robot hardware

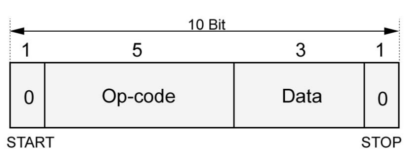

Figure 4. Structure of robot's messages

B. *Communication Module Design*

Each AMR's messages are crafted using similar packet format as shown in Fig. 4. The broadcast mode communication is used for obstacle detection and trajectory information sharing with neighboring AMRs. In this mode, all data broadcasted with the IR emitters, will be detected by the phototransistors.

Confirmation-based message communication are defined for complex cooperation of swarm multiple robots. The flowchart for the short-range communication technique is shown in Fig. 6. Confirmation-based message is used to define high-level behaviors in swarm mobile robots. In these scenarios, after receiving another AMR's message, the individual AMR will transmit its own message packet to initiate their connection and wait for an acknowledgement message during a defined time. The connection request will be eliminated if the AMR does not receive any acknowledgement response. Fig. 7 shows confirm-based messages sequence to start new connection between two AMRs.

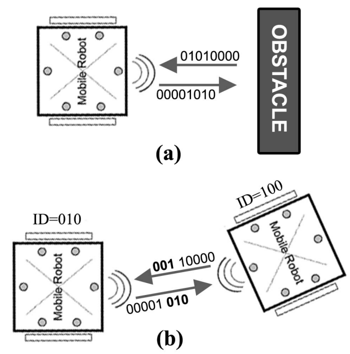

Figure 5. Detection principles: (a) robot detects the obstacle if receives its ID packet that was reflected, (b) recognizes other robot if receives new ID

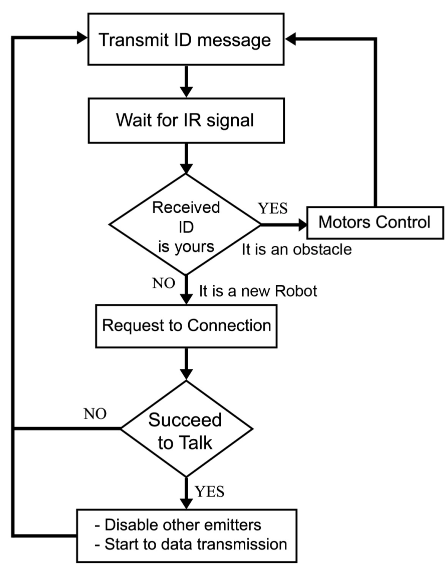

Figure 6. Flowchart of the developed data transmission technique

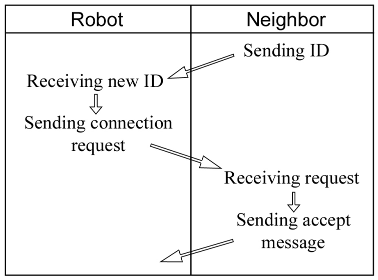

Figure 7. Confirm-based messages to start new connection

V. EXPERIMENTAL RESULTS

The proposed communication system was tested in several environments such as sunlight, dark room, and fluorescent light room. The sunlight includes infrared radiation and results are significantly different when there is sunlight in the environment. The fluorescent light wavelength is less than 700nm and as a result, the measured values in fluorescent lighted room are almost similar with measurement from dark room. Black body and white body obstacle have been used evaluate the distance estimation function. Table I illustrates measured values in dark room and indirect sunlight outdoor environment for black body and white body obstacle.

Fig. 8 (a) illustrates the converted ADC values of reflected IR radiation from white body obstacle in indoor and outdoor environments. The white body obstacle reflects more IR radiation than black body. The measured samples with black body are shown in Fig. 8 (b). Distance estimation would depend on measured IR samples which are reflected from obstacles. In this design, the maximum distance for obstacle detection is about 12cm with tolerance ±1cm. The sensitivity of sensors would also depend on emitter diode intensity which is relative to emitter diode quality and battery level.

TABLE I. CAPTURED REFLECTED VALUES FROM BLACK BODY AND WHITE BODY OBSTACLES IN TWO ENVIRONMENTS

Distance (cm)	Indoor Fluorescent light		Outdoor Indirect Sunlight	
	Black body Obstacle	*White body Obstacle*	*Black body Obstacle*	*White body Obstacle*
0.5	220	238	237	239
1	238	240	235	241
1.5	234	241	178	239
2	224	241	138	237
3	132	234	103	187
4	93	154	88	134
5	72	112	82	109
6	63	90	78	96
8	54	68	74	86
10	50	59	71	80
15	47	50	69	72
20	44	45	66	67

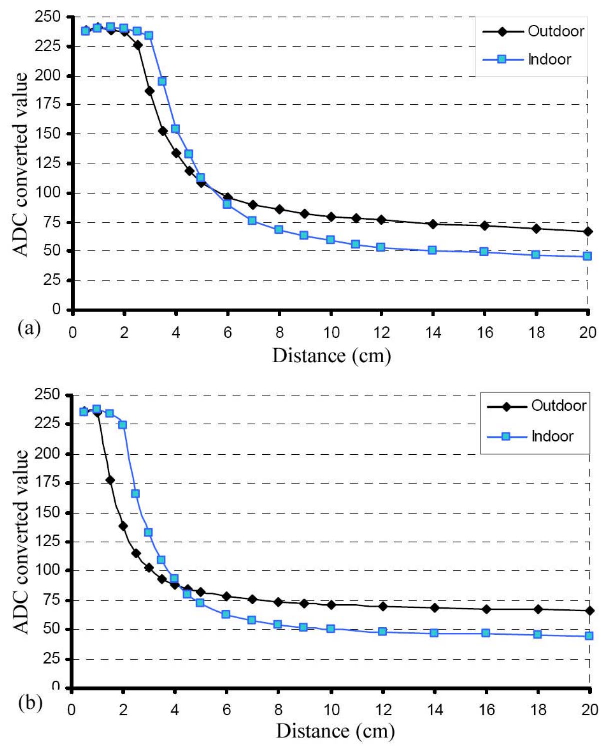

Figure 8. (a) ADC converted values from white body obstacle's reflection, (b) ADC converted values from black body obstacle's reflection

VI. CONCLUSION

This paper presents a short-range infrared communication technique for swarm mobile robots. A reliable short-range data communication has been designed using low-cost infrared components. The IR is used for obstacle detection, distance estimation and data communication in an autonomous mobile robot. Two type of communication mode are available to support the operation of AMRs. The broadcast communication mode allows sharing of trajectory information between neighboring robots while the confirmation-based message is used for reliable communication between AMRs. The proposed IR design has been evaluated using black body and white body object with promising result.

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